

1 Supplementary Material

2 1 Notations

Table 1: Summary of notation used throughout the paper

Symbol	Description	Dim./Type
$\mathcal{X} = \{x_i\}_{i=1}^n$	Full training dataset of n samples	set, $ \mathcal{X} = n$
x_i	i -th training example	$\mathbb{R}^d$
n	Number of training samples	scalar
d	Input-feature dimension	scalar
$X \in \mathbb{R}^{m \times n}$	Data matrix used in Fig. 1	matrix
B	Number of mini-batches per epoch	scalar
X_i	i -th mini-batch	subset of $\mathcal{X}$
K	Mini-batch size	scalar
T	Total training iterations	scalar
S	Subset-refresh interval	scalar
Θ	Trainable model parameters	vector / tensor
$L(\Theta; x)$	Loss on a single sample	scalar
$\nabla_{\Theta} L(\Theta; x)$	Per-sample gradient	$\mathbb{R}^{ \Theta }$
$\bar{g}_i$ or $\bar{g}$	Mean gradient of batch X_i	$\mathbb{R}^{ \Theta }$
g_r	Gradient of r -th selected sample	$\mathbb{R}^{ \Theta }$
$G \in \mathbb{R}^{d \times K}$	Gradient matrix of a full batch	matrix
$\mathbf{R} = \{R_1, \dots, R_c\}$	Candidate subset sizes (ranks)	set
R	A chosen rank / subset size	scalar
R^*	Rank minimising projection error	scalar
$\mathcal{S}_i^R$	Indices of R samples selected from X_i	index set
$\mathcal{S}_i$	Final index set for batch i after rank search	index set
$\mathcal{S}^t$	Aggregate subset used at iteration t	index set
$f(\cdot)$	Feature-extraction mapping (e.g. SVD encoder)	function
$\mathbf{V} = f(X_i) \in \mathbb{R}^{K \times R}$	Low-rank feature matrix for batch X_i	matrix
$\text{fastmax-volume}(\mathbf{V}, R)$	Fast MaxVol row-selection routine	algorithm
$\mathbf{G}_R$	Gradient matrix restricted to $\mathcal{S}_i^R$	$\mathbb{R}^{d \times R}$
d_R	Projection error $\ \bar{g} - \mathbf{G}_R \mathbf{G}_R^\dagger \bar{g}\ _2^2$	scalar
P_R	Orthogonal projector onto $\text{span}(\mathbf{G}_R)$	matrix
ϵ	Tolerance on gradient-projection error	scalar
σ_{R+1}	$(R+1)$ -st singular value of batch matrix	scalar
$p = [p_1, \dots, p_R]$	Row indices returned by MaxVol	vector
I	Number of batches aggregated at refresh	scalar
$\langle \hat{g}, g_R^t \rangle$	Cosine similarity between epoch and subset gradients	scalar
E	Estimated CO ₂ emissions	kg CO ₂
$\Phi_{\text{acc}}(E)$	Accuracy fidelity coefficient vs. emissions	function
$\Psi(f)$	Accuracy vs. trained data fraction f	function
λ, E_0, H	Parameters of the exponential efficiency curve	scalars

3 2 Extended Algorithm

Gradient-Aligned Sampling periodically (every S iterations) scans each training mini-batch to find the smallest subset of examples whose gradients closely span the full-batch gradient it first averages per-example gradients to obtain $\bar{g}_i$, then, for each candidate rank R_i , uses a fast max-volume routine on the batch's feature matrix to pick R_i exemplars, forms their gradient matrix, and computes the reconstruction error d_{R_i} ; the rank R^* that minimizes this error yields the chosen indices $\mathcal{S}_i$, and

9 aggregating over batches gives the active subset $\mathcal{S}^t$ for that period. In iterations between renewals,
 10 the method simply reuses the previous subset, and all model parameters are updated using only the
 11 currently active $\mathcal{S}^t$, thus retaining the stability of large-batch gradients while evaluating far fewer
 12 samples.

Algorithm 1: Training with Gradient-Aligned Sampling

Input: Training dataset $\mathcal{X}$, ranks $\mathbf{R} = \{R_i\}_{i=1}^C$, feature matrices $\mathbf{V}$, selection period S , and batch size K

Output: Sampled data subset indices $\mathcal{S}^t$

begin

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for  $t = 1, \dots, T$  do
  Stage 1: Subset Selection
  if  $t \bmod S == 0$  then
    for each batch  $\mathcal{X}_i \subset \mathcal{X}$  do
       $\bar{\mathbf{g}}_i \leftarrow \frac{1}{K} \sum_{k=1}^K \nabla_{\Theta} L(\Theta^t; \mathcal{X}_i(:, k))$ 
      for  $R_i \in \mathbf{R}$  do
         $\mathcal{S}_i^r = \text{fastmax-volume}(\mathbf{V}_i, R_i)$ 
         $\mathbf{G}_{R_i} = [\dots, \nabla_{\Theta} L(\Theta^t; \mathcal{X}_i(:, \mathcal{S}_i^r(p))), \dots]$ 
        Gradient error  $d_{R_i} = \|\bar{\mathbf{g}}_i - \mathbf{G}_{R_i}^{\dagger} \bar{\mathbf{g}}_i\|_2^2$ 
      Optimal rank  $R^* = \arg \min_{R_i} \{d_{R_i}\}_{i=1}^{|\mathbf{R}|}$ 
      Store the selected indices  $\mathcal{S}_i = \mathcal{S}_i^{R^*}$ 
    Aggregate all subsets  $\mathcal{S}^t = \{\mathcal{S}_i\}_{i=1}^I$ 
  else
    Retain the previous subset  $\mathcal{S}^t = \mathcal{S}^{t-1}$ 
  Stage 2: Model Update
  Update all model parameters using  $\mathcal{S}^t$ 

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13 3 Detailed Proofs

14 **Remark 1** (Gradient Approximation Guarantee). Let $A \in \mathbb{R}^{K \times M}$ be a batch matrix, and let $V =$
 15 $U_R \in \mathbb{R}^{K \times R}$ denote the matrix of the top- R left singular vectors of A . Let $S \subset [K]$, $|S| = R$, be the
 16 subset of rows selected by the MaxVol algorithm applied to V . If the gradient map $g(x) = \nabla_{\theta} L(\theta; x)$
 17 is L_g -Lipschitz continuous, then

$$\|\nabla_{\theta} L(\theta; A) - \nabla_{\theta} L(\theta; A(S, \cdot))\|_2 \leq \frac{K}{R} L_g \sigma_{R+1},$$

18 where σ_{R+1} is the $(R+1)$ -th singular value of A .

19 **Proof.** Consider the singular value decomposition (SVD) of A , given by $A = U \Sigma V^{\top}$. Denote by
 20 $A_R = U_R \Sigma_R V_R^{\top}$ the rank- R truncation of A , and let the residual matrix be $E := A - A_R$, so that
 21 $\|E\|_2 = \sigma_{R+1}$. By construction, A_R captures the dominant R -dimensional subspace of A .

22 Now, let $V = U_R \in \mathbb{R}^{K \times R}$ and apply the MaxVol algorithm to V . The algorithm selects a subset
 23 $S \subset [K]$ of size R such that the submatrix $M = V(S, :)$ is non-singular. Define the interpolation
 24 matrix $T := V M^{-1} \in \mathbb{R}^{K \times R}$. According to properties established in MaxVol theory Goreinov et al.
 25 2010, $V = T V(S, :)$, and the entries of T satisfy $\max_{i,j} |T_{ij}| \leq 1$ and $\|T_{i,:}\|_1 \leq R$ for all rows i ,
 26 with the additional property that $\sum_{i=1}^K T_{ij} = \frac{K}{R}$ for each column j .

27 Since A_R shares the same left singular vectors as V , these properties allow us to express any row i of
 28 A_R as a linear combination of the sampled rows:

$$A_R(i, :) = T_{i,:} A_R(S, :), \quad \forall i \in [K]. \quad (1)$$

29 Consider now the gradient map $g(x) = \nabla_{\theta} L(\theta; x)$, assumed to be L_g -Lipschitz. For each row i ,
 30 define the interpolated gradient $\hat{g}_i := \sum_{j \in S} T_{ij} g(A(j, :))$. Using the reconstruction property (1) and

the Lipschitz continuity, we obtain

$$\|g(A(i, :)) - \hat{g}_i\|_2 \leq L_g(1 + \|T_{i,:}\|_1)\|E(i, :)\|_2 \leq L_g(1 + R)\sigma_{R+1},$$

where we have used the bounds $\|T_{i,:}\|_1 \leq R$ and $\|E(i, :)\|_2 \leq \sigma_{R+1}$.

To bound the overall gradient approximation error, consider the average gradient over all rows, $\bar{g} := \frac{1}{K} \sum_{i=1}^K g(A(i, :))$, and the average over the selected subset, $g_S := \frac{1}{R} \sum_{j \in S} g(A(j, :))$. Expanding $\bar{g} - g_S$ in terms of the interpolated gradients and leveraging the properties of the interpolation matrix, we find that the terms involving T sum to zero, leaving

$$\|\bar{g} - g_S\|_2 \leq \frac{1}{K} \sum_{i=1}^K L_g(1 + R)\sigma_{R+1} \leq \frac{K}{R} L_g \sigma_{R+1},$$

where the final bound uses the fact that $K \geq R$ and $1 + R \leq 2R$ for practical choices of R .

Finally, recognizing that $\bar{g} = \nabla_\theta L(\theta; A)$ and $g_S = \nabla_\theta L(\theta; A(S, :))$ completes the proof:

$$\|\nabla_\theta L(\theta; A) - \nabla_\theta L(\theta; A(S, :))\|_2 \leq \frac{K}{R} L_g \sigma_{R+1}.$$

□

Theorem 1 (Convergence via Gradient-Aligned Subspace Sampling). Let $\bar{g} \in \mathbb{R}^d$ denote the average gradient over a batch of K samples, and let $G_R \in \mathbb{R}^{d \times R}$ be a collection of R gradients selected via MaxVol, spanning a full-rank subspace $\mathcal{S}_R$. If the projection error $\|\bar{g} - \text{Proj}_{\mathcal{S}_R}(\bar{g})\|_2 \leq \varepsilon$, then gradient descent using the projected direction converges to a stationary point under standard assumptions Bottou, Curtis, and Nocedal 2018.

Assumptions. Suppose the loss function L is C -smooth, i.e., for all Θ, Θ' ,

$$\|\nabla L(\Theta) - \nabla L(\Theta')\|_2 \leq C\|\Theta - \Theta'\|_2.$$

Assume also that the gradients are bounded, $\|\nabla L(\Theta)\|_2 \leq G$ for all Θ , and the projection error satisfies $\|\bar{g} - g_{\text{proj}}\|_2 \leq \varepsilon$, where $g_{\text{proj}} = \text{Proj}_{\mathcal{S}_R}(\bar{g})$.

Then, for the gradient descent updates $\Theta_{t+1} = \Theta_t - \eta g_{\text{proj}}^{(t)}$, the iterates converge to a stationary point where $\|\nabla L(\Theta)\|_2 \leq \varepsilon G$.

Proof. By C -smoothness of L , the loss after a gradient step satisfies

$$L(\Theta_{t+1}) \leq L(\Theta_t) - \eta \langle \nabla L(\Theta_t), g_{\text{proj}}^{(t)} \rangle + \frac{C\eta^2}{2} \|g_{\text{proj}}^{(t)}\|_2^2.$$

Since the projection error is at most ε , we have

$$\langle \nabla L(\Theta_t), g_{\text{proj}}^{(t)} \rangle = \langle \nabla L(\Theta_t), \bar{g}^{(t)} \rangle - \langle \nabla L(\Theta_t), \bar{g}^{(t)} - g_{\text{proj}}^{(t)} \rangle \geq \|\nabla L(\Theta_t)\|_2^2 - \varepsilon G,$$

where we use Cauchy-Schwarz and the gradient bound $\|\nabla L(\Theta_t)\|_2 \leq G$.

Plugging this estimate into the smoothness inequality and recalling $\|g_{\text{proj}}^{(t)}\|_2 \leq G$, we obtain

$$L(\Theta_{t+1}) \leq L(\Theta_t) - \eta \left(\|\nabla L(\Theta_t)\|_2^2 - \varepsilon G \right) + \frac{C\eta^2 G^2}{2}.$$

Summing this inequality over $t = 0, \dots, T-1$ and dividing by T gives

$$\frac{1}{T} \sum_{t=0}^{T-1} \|\nabla L(\Theta_t)\|_2^2 \leq \frac{L(\Theta_0) - L(\Theta^*)}{\eta T} + \varepsilon G + \frac{C\eta G^2}{2},$$

where $L(\Theta^*)$ denotes the minimum loss. Setting $\eta = 1/\sqrt{T}$ ensures the first and last terms vanish as $T \rightarrow \infty$, leaving the bound $\|\nabla L(\Theta_t)\|_2 \leq \varepsilon G$ in the limit. This establishes convergence to a stationary point determined by the projection error. □

58 **Lemma 1** (Projection Error Bound). Let $\tilde{G}_R$ be an orthonormal basis for G_R . Then

$$\|\bar{g} - \tilde{G}_R \tilde{G}_R^\top \bar{g}\|_2^2 = \|\bar{g}\|_2^2 \left(1 - \left\| \frac{\tilde{G}_R^\top \bar{g}}{\|\bar{g}\|_2} \right\|_2^2 \right).$$

59 *Proof.* Since $\tilde{G}_R$ is orthonormal, the projection of $\bar{g}$ onto $\text{span}(G_R)$ is given by

$$g_{\text{proj}} = \tilde{G}_R \tilde{G}_R^\top \bar{g}.$$

60 The residual $r = \bar{g} - g_{\text{proj}}$ is orthogonal to $\text{span}(G_R)$. By the Pythagorean theorem,

$$\|\bar{g}\|_2^2 = \|g_{\text{proj}}\|_2^2 + \|r\|_2^2.$$

61 Because $\tilde{G}_R$ is orthonormal, we have $\|g_{\text{proj}}\|_2 = \|\tilde{G}_R^\top \bar{g}\|_2$. Substituting this, we obtain

$$\|r\|_2^2 = \|\bar{g}\|_2^2 - \|\tilde{G}_R^\top \bar{g}\|_2^2 = \|\bar{g}\|_2^2 \left(1 - \left\| \frac{\tilde{G}_R^\top \bar{g}}{\|\bar{g}\|_2} \right\|_2^2 \right),$$

62 which completes the proof. $\square$

63 **Corollary 1** (Dynamic Rank Adjustment Ensures Convergence). If the rank R is dynamically adjusted
64 so that $\|\bar{g} - \tilde{G}_R \tilde{G}_R^\top \bar{g}\|_2^2 \leq \varepsilon$ at every iteration, then GRAFT ensures convergence to a local minimum.

65 *Proof.* The result follows directly from Theorem 1, since the gradient approximation error remains
66 bounded by ε at each step, independent of changes in R . Specifically, increasing R refines the
67 subspace $\text{span}(G_R)$ and reduces ε , while decreasing R is permissible only if the new error still
68 satisfies $\varepsilon \leq \varepsilon_{\text{target}}$. In both scenarios, the condition $\|\bar{g} - g_{\text{proj}}\|_2 \leq \varepsilon$ is maintained, thereby
69 ensuring convergence. $\square$

70 4 Complexity Analysis of GRAFT

71 The computational complexity of the GRAFT algorithm can be analyzed by decomposing its major
72 operations into four principal components: feature extraction, sample selection, gradient computation,
73 and alignment evaluation.

74 **Feature Extraction (One-Time Cost).** Prior to entering the main training loop, GRAFT projects
75 each batch $X \in \mathbb{R}^{K \times d}$ into a lower-dimensional subspace, typically using SVD or a related feature
76 extraction method. The complexity of this initial projection is

$$\mathcal{O}(KdR), \tag{2}$$

77 where K is the batch size, d is the input dimension, and R is the reduced subspace dimension. This
78 operation is performed once per batch and does not contribute to the per-iteration computational cost.

79 **Fast MaxVol Selection (Per Iteration).** At each iteration, GRAFT employs the Fast MaxVol
80 algorithm to select a subset of R samples from the batch of size K . The complexity of this selection
81 step is

$$\mathcal{O}(KR^2), \tag{3}$$

82 which is incurred at every iteration and scales with both the batch size and the target subset size.

83 **Gradient Computation (Per Iteration).** Once the subset is selected, gradients are computed only
84 for the selected R samples. The corresponding per-iteration complexity is

$$\mathcal{O}(Rd), \tag{4}$$

85 which remains efficient due to the linear scaling with respect to the subset size rather than the full
86 batch.

87 **Alignment Checks (Per Iteration).** To ensure adequate gradient alignment, GRAFT evaluates the
 88 alignment quality across a predefined set of candidate ranks $|R_{\text{set}}|$. The complexity associated with
 89 these alignment checks at each iteration is

$$\mathcal{O}(|R_{\text{set}}| \cdot Rd), \quad (5)$$

90 where $|R_{\text{set}}|$ is the number of rank candidates evaluated.

91 Combining all per-iteration components, the total complexity per training iteration is given by

$$\mathcal{O}(KR^2 + Rd + |R_{\text{set}}| \cdot Rd). \quad (6)$$

92 This expression reflects the cumulative cost of subset selection, gradient computation, and rank-
 93 based alignment evaluation, with the one-time feature extraction cost excluded from per-iteration
 94 considerations.

Method	Batch-wise	Gradient-based	Complexity	Scalability	Memory Usage
DRoP Vysogorets, Ahuja, and Kempe 2025	✗	✗	$\mathcal{O}(n^2d)$	✗	High
GradMatch Killamsetty et al. 2021	✓	✓	$\mathcal{O}(nkd)$	✗	High
CRAIG Mirzasoleiman, Bilmes, and Leskovec 2020	✓	✓	$\mathcal{O}(nkd)$	✗	Moderate
GRAFT	✓	✓	$\mathcal{O}(KR^2 + R_{\text{set}} \cdot Rd)$	✓	Low

Table 2: Complexity comparison of subset selection methods. Here, n is the dataset size, d is the feature dimension, K is the batch size, R is the subset size, and $|R_{\text{set}}|$ is the number of predefined rank candidates.

95 4.1 Additional Experimental Details

96 In this supplementary section, we provide a comprehensive set of additional experiments to further
 97 assess the generalizability and effectiveness of the proposed sampling approach across diverse
 98 machine learning tasks. First, we evaluate our method in the context of fine-tuning large language
 99 models (LLMs), demonstrating its scalability and impact in high-dimensional settings. We also
 100 include results on classical regression benchmarks to illustrate the method’s applicability beyond
 101 deep learning. Beyond empirical results, we investigate the convexity properties of the proposed
 102 sampling technique, analyzing how the sample selection mechanism influences the underlying loss
 103 landscape. This analysis provides theoretical justification for the observed stability and convergence
 104 behavior in our experiments. To offer a fair and transparent evaluation, we compare our approach
 105 against several widely used training and sampling strategies, including random sampling, core-set
 106 methods, and recent gradient-based subset selection algorithms. We report the standard performance
 107 metrics emissions by each method.

108 Collectively, these extended experiments and analyses aim to provide a deeper understanding of the
 109 strengths and potential limitations of the proposed sampling framework.

110 4.2 Experimental Details on Fine-Tuning BERT

111 In this experiment, we apply the proposed sampling method to the fine-tuning of large language
 112 models (LLMs), with a focus on practical NLP tasks. Specifically, we evaluate our approach
 113 on the IMDB sentiment analysis dataset by fine-tuning a distilled variant of BERT containing
 114 approximately 92 million parameters. Our sampling strategy, GRAFT, was executed on the embedding
 115 representations generated by the language transformer, enabling efficient selection of informative
 116 samples for gradient computation. The IMDB dataset is a well-established benchmark for text
 117 analytics, comprising 50,000 movie reviews labeled by sentiment. The dataset is split evenly, with
 118 25,000 reviews allocated for training and 25,000 for testing. This setup allows us to thoroughly
 119 evaluate the generalization performance and efficiency of our sampling approach in a real-world NLP
 120 context.

121 **Experimental Setting.** For fine-tuning, we employed a batch size of 100, a constant learning rate
 122 of 5×10^{-5} , and a weight decay of 0.0001. The BERT model was fine-tuned for 30 epochs under two
 123 data regimes: the full dataset, and a GRAFT-selected subset generated using our GRAFT algorithm.
 124 For GRAFT, both standard and warm-start variants were tested, with data subset selection performed
 125 every 10 epochs. To assess the effect of sample efficiency, only 35% of the original training data was
 126 retained for each GRAFT selection cycle.

4.3 Extended Details and Experiments on Different Fractions of Data

In this section, we systematically compare the performance of various subset selection methods across different subset sizes (5%, 15%, 25%, and 35%). To ensure a fair comparison, we utilized the ResNeXt29_32x4d architecture Xie et al. 2017 for all datasets, training each model for 200 epochs, except for TinyImageNet and Caltech256, where a ResNet-18 model was used due to dataset-specific considerations. All experiments were conducted with a fixed batch size of 200 and an initial learning rate of 0.1, using stochastic gradient descent (SGD) as the optimizer. For TinyImageNet and Caltech256, a smaller batch size of 100 was employed. A CosineAnnealing learning rate scheduler was consistently applied to adjust the learning rate during training. All models were trained from scratch on an NVIDIA Tesla V100-SXM2 GPU (16GB) with an Intel(R) Xeon(R) Gold CPU, except for TinyImageNet and Caltech256, which were trained on an NVIDIA A100-SXM4 GPU (40GB).

Power consumption and CO₂ emissions were estimated using the methodology and tooling from Budennyy et al. Budennyy et al. 2022. Specifically, following the eco2AI framework, the instantaneous power usage of the computing hardware (GPU and CPU) is monitored throughout the training process. The framework records the real-time power draw (in watts) using hardware sensors or system queries, integrating these measurements over the training duration to obtain the total energy consumption (in kilowatt-hours, kWh) as

$$E = \frac{1}{3600} \sum_{i=1}^N P_i \Delta t_i, \quad (7)$$

where P_i is the instantaneous power at time step i , Δt_i is the time interval since the last measurement, and N is the total number of intervals. *Note:* In our main text, we report CO₂ emissions using the standard formula $\mathcal{E} = P \times t \times I$, which assumes a constant average power P over training duration t . The eco2AI library Budennyy et al. 2022 also supports a more granular mode based on real-time power monitoring and numerical integration, yielding equivalent results in our single-GPU training regime. To convert total energy consumption into CO₂ emissions, eco2AI multiplies E by a region-specific carbon intensity factor C (kg CO₂ per kWh), typically reflecting the local energy mix. The resulting emissions are calculated as

$$\text{CO}_2 = E \cdot C, \quad (8)$$

where C is set according to the geographic location of the compute cluster or is selected from published averages (for example, Germany: $C = 0.366$ kg CO₂/kWh). This approach enables standardized and transparent reporting of environmental impact for machine learning experiments.

Comparison on CIFAR-10 Dataset On CIFAR-10, 3 all subset selection methods achieve significant reductions in CO₂ emissions compared to full-data training (0.2192 kg). GRAFT yields the lowest emissions across all fractions, dropping to just 0.0656 kg at 25% and 0.0231 kg at 5%, but with a corresponding trade-off in accuracy (73.36% at 25% and 0.0231 kg at 5%, but with a corresponding trade-off in accuracy (73.36% at 25%). Notably, GRAFT Warm attains near-full accuracy even at reduced subsets, achieving 91.28% at 25% and 90.86% at 5%, while keeping emissions much lower than full-data training. Other methods like GLISTER, CRAIG, and GradMatch show similar trends, with emissions and accuracies falling between GRAFT and GRAFT Warm. DRoP, though efficient in terms of emissions, exhibits a steep drop in accuracy, especially at lower fractions. Overall, GRAFT Warm offers the best balance between environmental savings and model performance on CIFAR-10.

Table 3: CIFAR-10: Training Methods Comparison

Method	0.05		0.15		0.25		0.35	
	CO ₂	Acc.	CO ₂	Acc.	CO ₂	Acc.	CO ₂	Acc.
Full	0.2192	93.21	0.2192	93.21	0.2192	93.21	0.2192	93.21
GRAFT	0.0231	73.36	0.0454	86.02	0.0656	88.87	0.0828	91.74
GRAFT Warm	0.0480	90.86	0.0688	91.41	0.0822	91.28	0.0938	92.49
GLISTER	0.0584	74.96	0.0725	86.60	0.0938	90.22	0.0846	91.64
CRAIG	0.0547	73.36	0.0802	86.24	0.0935	88.60	0.0884	90.58
GRADMATCH	0.0549	75.26	0.0658	85.98	0.0793	89.43	0.0854	91.66
DRoP	0.0372	46.12	0.0428	67.35	0.0471	75.21	0.053	81.50

Comparison on CIFAR-100 dataset For CIFAR-100, the pattern of CO₂ and accuracy trade-offs remains consistent. GRAFT achieves the lowest emissions (0.0240 kg at 5%), but its accuracy at this extreme fraction is substantially reduced (40.60%). GRAFT Warm, meanwhile, maintains a higher accuracy (70.82% at 25% subset and 68.20% at 5%) while still realizing considerable emissions savings relative to the full dataset (0.2212 kg, 75.45%). Other approaches GLISTER, CRAIG, and GradMatch achieve intermediate performance, with emissions and accuracy reflecting their respective sample selection strategies. DRoP minimizes emissions most aggressively but with pronounced accuracy loss. These results indicate that GRAFT Warm consistently provides the best compromise on CIFAR-100, combining robust performance with substantially reduced environmental impact.

Table 4: CIFAR-100: Training Methods Comparison

Method	0.05		0.15		0.25		0.35	
	CO ₂	Acc.	CO ₂	Acc.	CO ₂	Acc.	CO ₂	Acc.
Full	0.2212	75.45	0.2212	75.45	0.2212	75.45	0.2212	75.45
GRAFT	0.0240	40.60	0.0477	60.50	0.0763	64.50	0.1054	73.52
GRAFT Warm	0.0496	68.20	0.0707	69.31	0.0895	70.82	0.1138	73.63
GLISTER	0.0626	26.70	0.0812	47.10	0.0755	48.70	0.0863	70.48
CRAIG	0.0633	24.80	0.0678	42.40	0.1011	48.50	0.0799	66.56
GRADMATCH	0.0553	25.71	0.0682	40.70	0.0750	50.60	0.0873	70.44
DRoP	0.038	13.51	0.0460	24.11	0.050	31.32	0.056	38.20

Comparison on TinyImagenet Dataset On TinyImageNet Fei-Fei et al. 2015, all subset selection methods achieve considerable reductions in CO₂ emissions relative to full-data training (0.297 kg), but with varying impacts on accuracy. GRAFT demonstrates the most favorable balance, reducing emissions to 0.092 kg at the 25% subset and achieving an accuracy of 0.545, closely approaching the full-data accuracy (0.590) at a fraction of the emissions. GRAFT Warm maintains high accuracy at 0.560 for 25% but with emissions comparable to the full-data case, indicating less efficiency in emission reduction. Other subset methods like CRAIG, GLISTER, and GradMatch achieve moderate emission savings but at the cost of greater accuracy drops compared to GRAFT. DRoP yields the lowest emissions at 0.048 kg for 25%, but with a significant reduction in accuracy (0.370). Overall, GRAFT provides the most effective trade-off between environmental impact and model performance on TinyImageNet.

Table 5: Comparison on TinyImageNet

Method	0.05		0.15		0.25	
	Emiss	Acc.	Emiss	Acc.	Emiss	Acc.
Full	0.297	0.590	0.297	0.590	0.297	0.590
DROP	0.052	0.130	0.047	0.261	0.048	0.370
GLISTER	0.094	0.267	0.140	0.471	0.210	0.517
CRAIG	0.097	0.248	0.170	0.424	0.220	0.485
GRADMATCH	0.093	0.257	0.168	0.507	0.216	0.556
GRAFT	0.049	0.406	0.053	0.505	0.092	0.545
GRAFT Warm	0.092	0.482	0.163	0.493	0.221	0.560

Comparison on Caltech256 Dataset Table 6 summarizes the trade-offs between CO₂ emissions and test accuracy for subset selection methods on the Caltech256 Griffin, Holub, and Perona 2007 dataset with ResNet18 Krizhevsky, Hinton, et al. 2009. All subset selection approaches yield substantial reductions in CO₂ emissions compared to full-data training (0.288 kg), with GRAFT achieving the lowest emissions at the 5% subset (0.049 kg). As expected, accuracy decreases as the subset fraction is reduced; however, GRAFT Warm achieves the highest accuracy among subset-based methods, attaining 0.60 at 25% and 0.55 at 5%, closely approaching the full-data accuracy (0.631) but with less than half the emissions.

Across all subset sizes, GRAFT and GRAFT Warm consistently outperform DRoP, GLISTER, CRAIG, and GradMatch in balancing emission reduction and accuracy. For example, at 25%, GRAFT Warm achieves 0.60 accuracy with 0.112 kg CO₂, whereas other methods yield lower

Table 6: Comparison on Caltech256

Method	0.05		0.15		0.25	
	Emiss	Acc.	Emiss	Acc.	Emiss	Acc.
Full	0.288	0.631	0.288	0.631	0.288	0.631
DROP	0.105	0.10	0.155	0.22	0.203	0.33
GLISTER	0.070	0.19	0.090	0.36	0.110	0.45
CRAIG	0.070	0.19	0.107	0.33	0.120	0.39
GRADMATCH	0.069	0.20	0.090	0.36	0.110	0.44
GRAFT	0.049	0.24	0.079	0.40	0.102	0.47
GRAFT Warm	0.0732	0.55	0.0981	0.58	0.112	0.60

accuracies or higher emissions. At more aggressive reductions (5% subset), GRAFT still provides a competitive 0.24 accuracy at only 0.049 kg emissions, and GRAFT Warm attains 0.55 accuracy with 0.073 kg emissions. These results demonstrate that GRAFT and its warm variant offer the best trade-off between environmental impact and predictive performance on Caltech256.

Performance comparison on FashionMNIST Dataset Table 7 presents a comparative analysis of training methods on the FashionMNIST dataset, focusing on data fraction, CO₂ emissions, and test accuracy. Full-data training serves as the reference, achieving 93.53% accuracy with 0.2118 kg of CO₂ emissions. Among subset-based approaches, GRAFT demonstrates high efficiency: at just 5% of the data, it achieves 88.76% accuracy while emitting only 0.0192 kg CO₂; at 35% data, it surpasses the full-data baseline in accuracy (93.74%) with less than half the CO₂ emissions (0.0779 kg). The GRAFT Warm variant further improves accuracy, reaching 89.97% at 5% data and 92.95% at 25% data, with emissions remaining substantially lower than full-data training. By contrast, alternative methods such as GLISTER, CRAIG, and GRADMATCH generally require greater computational resources and result in higher emissions to achieve similar accuracies. For example, GLISTER at 35% data yields 93.45% accuracy but with 0.0910 kg CO₂ emissions, which is notably higher than both GRAFT and GRAFT Warm. These results underscore the superior environmental and computational efficiency of GRAFT-based approaches for efficient model training on FashionMNIST.

Table 7: Comparison on FashionMNIST Dataset

Method	0.05		0.15		0.25		0.35	
	Emiss	Acc.	Emiss	Acc.	Emiss	Acc.	Emiss	Acc.
Full	0.2118	93.53	0.2118	93.53	0.2118	93.53	0.2118	93.53
GRAFT	0.0192	88.76	0.0396	91.66	0.0614	92.31	0.0779	93.74
GRAFT Warm	0.0195	89.97	0.0394	91.93	0.0614	92.95	0.0802	92.62
GLISTER	0.0611	89.62	0.0743	91.81	0.0861	92.88	0.0910	93.45
CRAIG	0.0633	89.87	0.0678	91.48	0.0832	92.52	0.0940	93.02
GRADMATCH	0.0563	88.33	0.0733	91.15	0.0860	92.43	0.0887	93.15
DRoP	0.042	81.01	0.0505	88.01	0.056	90.01	0.0633	91.25

GRAFT performance on medical dataset In this section, we extend our experiments to demonstrate the efficacy of our GRAFT methods on medical datasets. We selected the Dermamnist dataset from the MedMNIST collection and conducted training using various data fractions. Table 8 shows that using only 35% of the full dataset yields results comparable to using the entire dataset (74.06% vs. 76.06%), while training with 25% of the data achieves a validation accuracy of 73.47% and saves 35% of the training time, with reduced power consumption and CO₂ emissions. These results indicate that our GRAFT methods are effective for medical datasets as well.

Comparison of GRAFT and GRAFT Warm on Random sampling In table 9 we show the comparisons of our methods against random sampling. For random sampling we iteratively sample random subsets at every 25 epochs. In the table, we observe that both our methods outperform random sampling in terms of accuracy. Moreover, our GRAFT Warm has a better accuracy-efficiency tradeoff compared to random sampling.

Table 8: DermaMNIST Dataset

Method	0.05		0.15		0.25		0.35	
	Emiss	Acc.	Emiss	Acc.	Emiss	Acc.	Emiss	Acc.
Full	0.046	76.06	0.046	76.06	0.046	76.06	0.046	76.06
GRAFT	0.0093	67.78	0.0156	71.82	0.0217	73.47	0.0249	74.06

Table 9: CIFAR-10: Random Sampling and GRAFT Comparison

Method	0.05		0.15		0.25		0.35	
	Emiss	Acc.	Emiss	Acc.	Emiss	Acc.	Emiss	Acc.
Random	0.022	73.15	0.0459	85.73	0.0671	88.21	0.0902	90.21
GRAFT	0.0231	73.36	0.0454	86.02	0.0656	88.87	0.0828	91.74
GRAFT Warm	0.0280	90.86	0.0688	91.41	0.0822	91.28	0.0938	92.49

5 Impact of GRAFT Subset Training on the Loss Landscape

Understanding how data subset selection influences the loss landscape is crucial for interpreting model generalization and optimization behavior. Prior studies, such as Li et al. 2018, have shown that architectural innovations (e.g., skip connections) and training hyperparameters (e.g., batch size, learning rate, optimizer choice) can substantially affect the geometry of the loss surface. Building on these insights, we investigate how GRAFT-based subset training modifies the loss landscape relative to conventional full-data training. While a comprehensive theoretical analysis of data pruning and landscape geometry is beyond the present scope, we provide empirical evidence and defer in-depth investigation to future work.

Experimental Setup. We trained a MobileNetV2 Sandler et al. 2018 model on CIFAR-10 for 200 epochs using stochastic gradient descent (SGD) with Nesterov momentum, a batch size of 200, and weight decay of 0.0001. The learning rate was scheduled with CosineAnnealing, with a maximum of 0.01. To assess the effects of subset selection, we visualize the loss landscape as both 3D surface plots and 2D contour plots. This visualization facilitates direct evaluation of sharpness and flatness in the loss basin.

Figure 1 compares the loss surfaces for models trained on the complete dataset versus those trained with GRAFT-selected subsets. The left panels display the contour maps, while the right panels show 3D surfaces. Our results indicate that GRAFT subset training produces only minor perturbations to the overall loss geometry compared to full-data training. In both settings, the central minimizer exhibits sharpness and is surrounded by predominantly convex contours, with minimal evidence of non-convexity or irregular structure. Notably, the minimizer found using the GRAFT subset remains comparably sharp and well-localized, suggesting that GRAFT preserves a favorable loss landscape similar to full-data training.

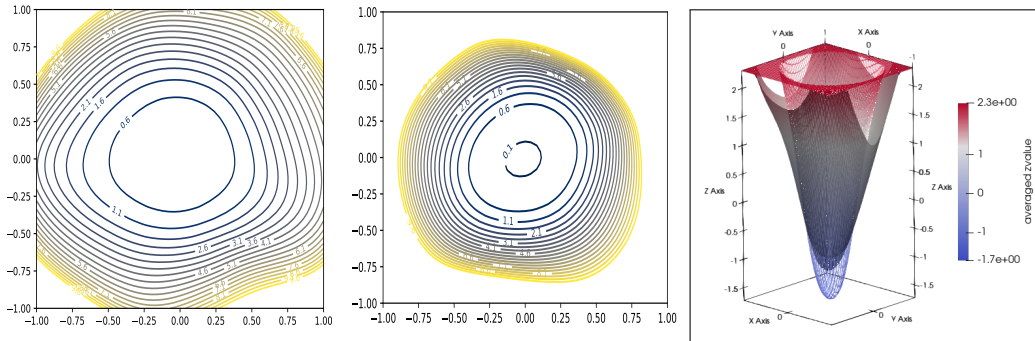


Figure 1: Effects on Loss Landscape

6 Limitations

Despite its appealing efficiency-accuracy trade-offs, GRAFT has several noteworthy limitations, its success critically depends on the quality of the chosen feature extractor, as the Fast MaxVol selector works on low-rank feature embeddings, and if the projection fails to capture relevant features, the selected subset will not effectively span the full-batch gradient, leading to degraded convergence. Furthermore, while GRAFT's dynamic adjustment mechanism aims to preserve gradient alignment, the algorithm can be sensitive to mini-batch size and class balance, as very small or highly skewed batches may cause unstable selections that no longer reliably represent the global gradient direction. Lastly, performance deteriorates under extreme compression; for example, selecting only 5% of CIFAR-100 data reduces top-1 accuracy from roughly 75% to 41%, demonstrating a clear limit to how aggressively data can be pruned.

7 Details on Datasets

CIFAR-10: The CIFAR-10 dataset comprises 60,000 colorful images, each with dimensions of 32×32 pixels, distributed across 10 distinct categories. Within each category, there are precisely 6,000 images, contributing to a balanced distribution. This dataset is segregated into five training sets and one test set, with each set containing 10,000 images. Specifically, the test set is composed of precisely 1,000 randomly selected images from each category, ensuring representation across classes.

CIFAR-100: Similar to CIFAR-10, the CIFAR-100 dataset boasts 100 classes, each featuring 600 images. Within each class, there are 500 training images and 100 testing images, maintaining a balanced distribution for robust model training and evaluation. However, CIFAR-100 introduces a hierarchical structure by grouping the 100 classes into 20 superclasses. Every image in this dataset is assigned two labels: a "fine" label indicating its specific class and a "coarse" label denoting its superclass.

Fashion-MNIST: Fashion-MNIST, an image dataset by Zalando, comprises 60,000 training examples and 10,000 test examples, each depicting grayscale images sized 28×28 . These images are classified into 10 distinct categories. Zalando's aim is to offer Fashion-MNIST as a direct substitute for the original MNIST dataset, enabling seamless benchmarking of machine learning algorithms. Notably, Fashion-MNIST maintains the same image dimensions and split structure as MNIST, facilitating effortless comparisons between models.

Dermamnist: The Dermamnist dataset is part of the MedMNIST collection and is designed for the classification of skin lesion images. It contains a diverse set of images representing various skin conditions, annotated with corresponding labels for seven different dermatological diseases. The dataset is intended for use in medical image analysis and machine learning research, providing a benchmark for evaluating models in the context of dermatology.

8 Feature selections models

In this section we introduce different feature selection methods such as SVD, ICA, PCA and Convolutional Neural Network (CNN) based Encoders.

- **Singular Value Decomposition (SVD)** Singular value decomposition (SVD) is a method of representing a matrix as a series of linear approximations that expose the underlying meaning-structure of the matrix. The goal of SVD is to find the optimal set of factors that best predict the outcome. SVD has been used to find the underlying meaning of terms in various documents. SVD reduces the overall dimensionality of the input matrix to a lower dimensional space (a matrix of much smaller size with fewer variables), where each consecutive dimension represents the largest degree of variability (between terms and documents) possible Miner et al. 2012.
- **Independent Component Analysis (ICA)** Independent Component Analysis (ICA) is a statistical method used to identify hidden factors of random variables. It is a linear generative model which assumes the observed variables are a linear mixture of unknown

non Gaussian and mutually independent variables. The aim of ICA is to find those variables without making any assumptions about the mixing system. More formally, if the data are represented by the vector $x = (x_1, \dots, x_n)$ and the independent component by the vector $s = (s_1, \dots, s_n)$, the aim of ICA is to find a linear transformation W verifying $s = Wx$ and minimizing a function F measuring the statistical independence Hotel et al. 2018

- **Principle Component Analysis PCA** Principal component analysis (PCA) is one of the most widely used multivariate analysis techniques. The aim of PCA is to reduce the data to a few characteristic dimensions for visualisation and analysis. This is achieved by calculating a new set of variables, or principal components, each of which is a linear combination of the original variables. The principal components (PCs) are chosen so that the important information in the data is retained in just a few of these new variables, effectively summarising the samples or observations. Better approximations are obtained by using more PCs, where each successive PC is uncorrelated with the previous PCs and expresses as much of the remaining variance as possible. This is achieved from the eigenvalue decomposition of the data covariance matrix with the coefficients for the k 'th principal component determined by the eigenvector of the covariance matrix corresponding to the k th largest eigenvalue. Data reduction is achieved by only keeping the first few PCs, which contain most of the information in the data McKenzie et al. 2011.
- **CNN based Encoders** Convolutional autoencoder extends the basic structure of the simple autoencoder by replacing the fully connected layers to convolution layers. Encoder use convolutions and decoders use transposed convolutional layers. CNN based encoders can reduce the less important feature coefficients to zero and the magnitude of the coefficients can be used as feature scores.

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Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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